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When Is Conformal Coverage Free? Switching Thresholds for Predict-then-Optimize

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Abstract

Machine learning increasingly drives operational decisions (dispatching power plants, routing vehicles, allocating medical supplies), yet its forecasts are uncertain. Conformal prediction turns a forecast into a calibrated uncertainty set with distribution-free guarantees, and a robust optimizer can then hedge its decision against that set. Before adopting this, a practitioner wants to know: will maintaining the uncertainty set actually change the decision the system makes, or will it leave the decision untouched and merely add an audit trail? We answer with a single quantity, the *switching threshold*: the point at which calibrated uncertainty begins to alter the chosen action. While the uncertainty stays below this threshold, coverage is *free*, and the system gains calibrated, auditable uncertainty without changing what it does. We show how to read this threshold from the structure of the decision problem, characterize the fluctuations of the online quantile that decide which side of it a problem falls on, and reduce these to a single pre-deployment *safety margin* that labels a problem as free, borderline, or costly. The label is set by the decision problem's structure rather than the choice of predictor. Across real benchmarks in energy, routing, public health, and logistics, dispatching power under real market prices is free (coverage never changes which generators run and adds no cost), whereas routing on a dense city map changes the route about half the time. The certificate stays reliable at city scale, on road networks with tens of thousands of streets, and even when the predictor is a vision model reading images. The result is a simple test for whether adding conformal coverage will cost a decision system anything.

Keywords: conformal prediction, uncertainty quantification, AI-driven optimization, decision-neutral coverage, switching threshold

1. Introduction

Operational AI systems routinely make consequential decisions under uncertainty.[1] Energy operators dispatch generators from demand forecasts, logistics platforms route vehicles from predicted travel times, and public health agencies allocate antivirals from epidemiological models. In each case, an ML model produces a prediction and a downstream optimizer acts on it. The prediction is never perfect, and in many domains the cost of being wrong is asymmetric: under-procurement of energy reserves incurs real-time penalty multipliers of 1.5–10× (Renkema et al., 2024; Shen et al., 2026), while under-allocation of antivirals during an epidemic is costlier than over-allocation.

1. Code: `https://github.com/chandrad/conformal-ops` (`pip install conformal-ops`).

Conformal prediction (Vovk et al., 2005) quantifies this uncertainty with distribution-free coverage guarantees: the true outcome falls within the prediction set at a user-specified rate $1 - \alpha$. In nonstationary settings where exchangeability fails, Adaptive Conformal Inference (ACI) (Gibbs and Candès, 2021) provides a complementary mechanism: a feedback controller that adjusts the confidence level of any set predictor to track the target miscoverage rate online. In modern AI pipelines, these calibrated prediction sets feed directly into a downstream optimizer that hedges against the uncertainty region.

This paper asks a simple question: *does the prediction set actually change the optimal decision?* If the set is small relative to the "gap" between alternative decisions, the optimizer selects the same solution whether it hedges or not, and conformal coverage is *free*. If the set is large enough to close the gap, the optimizer switches to a more conservative solution, and coverage has a price.

The question is practical. An energy company considering conformal coverage for its dispatch system wants to know *before deployment* whether the coverage certificate will change its operations or simply provide an audit trail at no cost. A logistics company routing thousands of vehicles wants to know whether conformal uncertainty sets will reroute traffic or leave existing routes unchanged.

We formalize this question and give a complete answer for a broad class of optimization problems; Figure 1 summarizes the pipeline and the resulting free/critical/costly triage. Our main contributions:

1. **The switching threshold** (Theorems 2–3): a single quantity, written κ^*, that is the level of conformal uncertainty at which the set first becomes large enough to close the gap between competing decisions described above. Below it coverage is *free* (the optimizer keeps its decision); above it coverage has a price. For a broad, well-structured class of problems, namely totally unimodular (TU) linear programs such as shortest-path, network-flow, and assignment, this boundary is *exact*. Its analytical heart is not the threshold itself but a characterization of how the online conformal uncertainty *fluctuates* around it, and how the threshold shrinks as problems grow.
2. **A pre-deployment safety margin** (Theorems 5–6): a single number, written m and read off a short calibration run, that sorts a problem into *free*, *critical*, or *costly*. It follows from characterizing how the online (Gibbs–Candès/ACI) conformal quantile fluctuates in steady state.
3. **Dimension dependence** (Proposition 7): the threshold shrinks in proportion to $1/K_{\text{det}}$, where K_{det} counts the near-optimal alternative decisions (for routing, the number of competing detour routes). This explains why small problems enjoy free coverage while large, dense ones do not.
4. **Validation on four real-world benchmarks** spanning energy, transportation, public health, and logistics, plus two case studies that stress-test the framework at scale: five real road networks with up to $\sim$40,000 decision variables (the problem dimension d, here the number of road links) under validated equilibrium congestion, and a deep vision predictor (Warcraft II terrain routing). Each comes with an analysis of why it falls into its predicted regime.

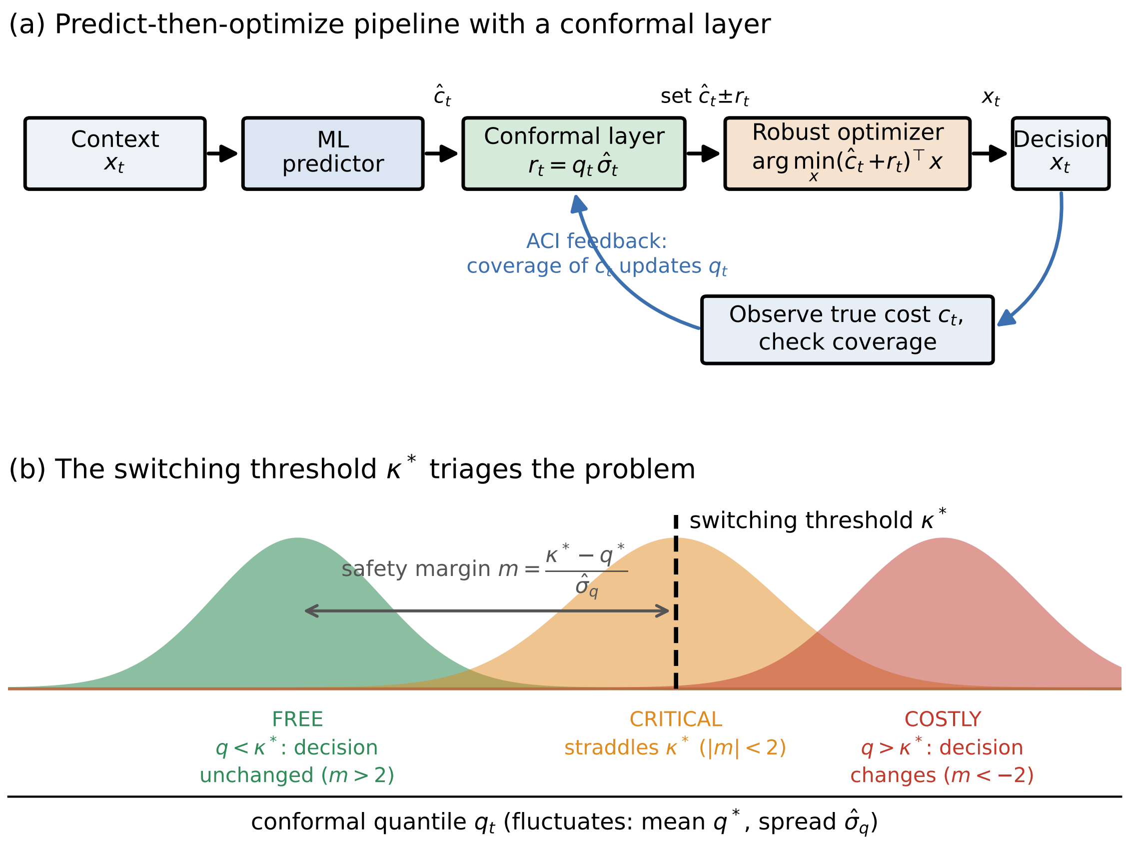

Figure 1: **Overview. (a)** A predict-then-optimize pipeline: a model predicts a cost vector $\hat{c}_t$, a conformal layer inflates it into an uncertainty set of radius $r_t = q_t\hat{\sigma}_t$, and a robust optimizer returns the decision; the ACI update adjusts the quantile q_t from realized coverage. The robust decision differs from the nominal one exactly when q_t exceeds the *switching threshold* κ^*. **(b)** κ^* is a structural property of the problem; the quantile q_t is a fluctuating data-driven quantity (mean q^*, spread $\hat{\sigma}_q$). Our safety margin $m = (\kappa^* - q^*)/\hat{\sigma}_q$ sorts the problem into *free* (coverage never changes the decision), *critical*, or *costly* (coverage routinely reroutes it).

2. Related Work

2.1. Conformal Prediction and Online Calibration

Conformal prediction, introduced by Vovk et al. (2005) within the framework of algorithmic randomness, provides prediction sets with finite-sample coverage guarantees under exchangeability. The theory has been extended considerably. Barber et al. (2023) relax exchangeability to weighted variants that accommodate certain forms of distribution shift. Feldman et al. (2023) connect conformal prediction to online learning, achieving risk control via sequential calibration. Angelopoulos et al. (2024) analyze the Gibbs-Candès update with decaying step sizes, establishing pointwise convergence of the miscoverage rate. Hallberg Szabadváry (2024) extend adaptive conformal inference to multi-step ahead time-series forecasting online, maintaining finite-sample coverage at each horizon even under non-exchangeability.

The Gibbs-Candès adaptive update (Gibbs and Candès, 2021) is central to our work. It tracks the target miscoverage rate α via stochastic approximation, providing long-run coverage at rate $O(\eta + 1/(\eta T))$ without distributional assumptions. Zaffran et al. (2022) aggregate over a range of step sizes to remove the learning-rate sensitivity, and Bhatnagar et al. (2023) obtain strongly-adaptive regret guarantees; Gibbs et al. (2025) extend the update to conditional guarantees, and Ramalingam et al. (2025) connect it to no-regret learning. Our contribution to this line of work is a characterization of quantile *fluctuations* (Theorem 5), going beyond convergence to the stationary variance, and connecting these fluctuations to polytope geometry via the switching threshold κ^*.

2.2. Conformal Prediction for Decision-Making

A growing body of work applies conformal prediction to downstream decision-making. The CPO framework (Patel et al., 2024) uses split conformal sets for batch robust optimization. Chenreddy and Delage (2024) train Conditional Robust Optimization end-to-end, learning the predictor and uncertainty set jointly. Yeh et al. (2025) go further by differentiating through the conformal calibration step to minimize downstream cost. Within the COPA community, Lanngren et al. (2025) compare conformal predictive distributions against Bayesian and point-predictive approaches for utility-maximizing decisions, finding that conformal methods trade some sharpness for formal coverage. Althoff et al. (2023) evaluate conformal-based probabilistic forecasting for wind speed prediction with normalized nonconformity scores, a setting closely related to our energy dispatch benchmark.

On the theoretical side, Kiyani et al. (2025) prove that prediction sets are optimal for VaR-optimizing agents, and Lekeufack et al. (2024) develop a conformal decision theory that calibrates prediction sets directly against downstream decision loss. Cortes-Gomez et al. (2025) tilt conformal scores toward utility-weighted losses. Wang and Dobriban (2026) characterize the minimax-optimal policy given a prediction set. Zhou and Zhu (2026) trace the miscoverage-regret Pareto frontier. Tang et al. (2025) apply calibrated uncertainty sets to route planning, closely related to our taxi routing benchmark.

These works all address how to *use* prediction sets in decisions: constructing better sets, optimizing decisions given a set, or trading off coverage against cost; Dronavajjala (2026) adapts the conformal prediction set to decision-relevant components in a healthcare ICU setting. Concretely, it reshapes the per-component uncertainty radii using the downstream allocation (tightening them on resources the optimizer barely uses, widening them where it allocates heavily) to cut the premium a robust optimizer pays for coverage while holding the target rate. We address a logically prior question: *when does the prediction set change the decision at all?* The switching threshold κ^* is derived from problem structure (computed from the constraint matrix and cost gaps), and is tight on TU polytopes. The substantive contribution is not the threshold itself, which is by definition the first switch point, but the quantile-fluctuation analysis and dimension dependence that determine whether a given problem operates above or below κ^*.

2.3. Predict-then-Optimize

The predict-then-optimize (PtO) paradigm (Elmachtoub and Grigas, 2022; Bertsimas and Kallus, 2020) uses ML predictions as inputs to optimization problems. SPO+ (Elmachtoub and Grigas, 2022) provides decision-focused surrogate losses that account for downstream optimization structure, and decision-focused learning more broadly trains the predictor through the optimization layer (Wilder et al., 2019). Bertsimas and Kallus (2020) develop Conditional Stochastic Optimization (CSO) using conditional distributions from nearest-neighbor methods. More recently, Capitaine et al. (2026) extend decision-focused learning to the online setting with regret bounds, and Im et al. (2025) extend surrogate losses to robust constraints with conformal sets. We add a conformal coverage layer to any fixed PtO pipeline and characterize when this layer is decision-neutral.

2.4. Robust Optimization and the Price of Robustness

Bertsimas and Sim (2004) define the *price of robustness* for budget-of-uncertainty sets: their price also vanishes for small budget Γ, analogous to our result for small q. Our setting differs in three ways: (i) the uncertainty set is conformal-calibrated (data-driven radius, not a worst-case budget parameter); (ii) the analysis is online with adaptive radii under distribution shift; and (iii) we provide an exact switching threshold κ^* as a function of polytope geometry, giving an if-and-only-if characterization rather than a sufficient condition. Distributionally robust optimization (Mohajerin Esfahani and Kuhn, 2018; Duchi and Namkoong, 2021) provides finite-sample guarantees under distributional ambiguity but cannot track coverage online under arbitrary shift.

3. Background

3.1. Conformal Prediction and Online Calibration

Conformal prediction (Vovk et al., 2005) constructs prediction sets with finite-sample coverage guarantees. Given a nonconformity score s_t and a target miscoverage rate $\alpha \in (0, 1)$, the prediction set $C_t = \{y : s(x_t, y) \leq q_t\}$ satisfies $P(y_t \in C_t) \geq 1-\alpha$ under exchangeability.

In the online setting with potential distribution shift, exchangeability may fail. Gibbs and Candès (2021) propose an adaptive update:

$$\alpha_{t+1} = \mathrm{clip}(\alpha_t + \eta(\alpha - \mathbf{1}[s_t > q_t]),\ \underline{\alpha},\ \bar{\alpha}) \tag{1}$$

where $\eta > 0$ is a learning rate and $q_t = Q_{1-\alpha_t}(\mathcal{S}_t)$ is the quantile from a sliding window $\mathcal{S}_t$ of recent scores. When the score exceeds the quantile (miscoverage), α_t increases, widening future prediction sets; when coverage succeeds, α_t decreases, tightening sets. This self-correcting mechanism drives long-run empirical coverage toward $1-\alpha$ (at rate $O(\eta+1/(\eta T))$ for the exact ACI update, without distributional assumptions); our sliding-window variant tracks this behavior, which we verify empirically on every benchmark. The iteration is a constant-step-size stochastic approximation (Borkar, 2008).

Relationship to conformal prediction. ACI is not itself a conformal predictor in the sense of Vovk et al. (2005); it is a feedback controller that can be composed with any confidence predictor. In our pipeline (Figure 2) the predictor is *inductive* (trained once

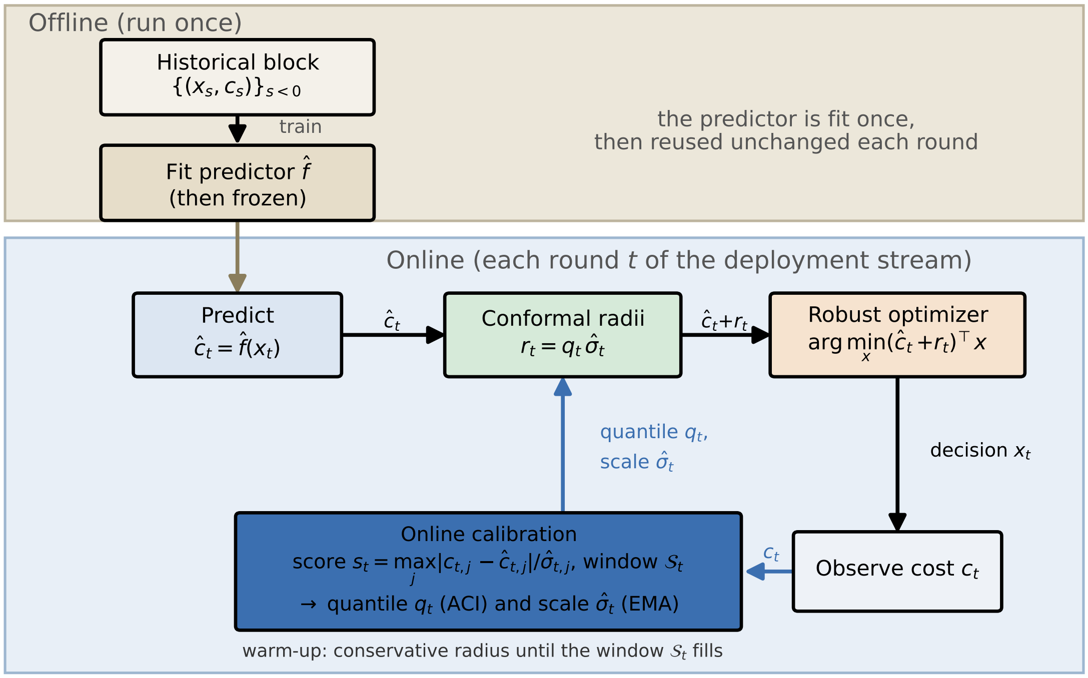

Figure 2: The CP architecture used throughout. **Offline (once):** the predictor $\hat{f}$ is fit on a historical block and then *frozen*. **Online (each round of the deployment stream):** $\hat{f}$ predicts the cost $\hat{c}_t$; conformal radii $r_t = q_t\hat{\sigma}_t$ inflate it; a robust optimizer returns the decision x_t; the true cost c_t is then revealed, turned into a nonconformity score s_t, and appended to a sliding window $\mathcal{S}_t$ of the most recent scores from which the ACI update maintains the quantile q_t. Calibration is thus *online* on the stream (with a conservative warm-up until $\mathcal{S}_t$ fills): an inductively-trained, fixed predictor with online ACI-driven calibration, not split conformal on a held-out calibration fold. The quantile q_t and scale $\hat{\sigma}_t$ used at round t depend only on scores from earlier rounds, so there is no look-ahead.

on a historical block, never retrained online), while coverage is calibrated *online*: at each round the per-component nonconformity score (2) is computed on the deployment stream, and the ACI update (1) maintains the quantile q_t from a sliding window of the most recent scores, defaulting to a conservative radius during an initial warm-up until enough scores have accumulated. This is online conformal calibration with a fixed predictor, following the train-on-history, evaluate-on-future protocol of all our experiments; it is *not* split-conformal calibration on a held-out fold. We refer to this composite of a fixed predictor and online, ACI-driven calibration as "conformal coverage" throughout.

3.2. Multivariate Conformal Scoring

When the predicted quantity is a d-dimensional cost vector $\mathbf{c}_t \in \mathbb{R}^d$, we need a conformity measure that produces a d-dimensional prediction set. A naive approach of independent conformal intervals per component requires a Bonferroni correction (setting per-component level α/d), which becomes overly conservative for large d. We avoid this by using the normalized max-residual score:

$$s_t = \max_j |c_{t,j} - \hat{c}_{t,j}|/\hat{\sigma}_{t,j} \tag{2}$$

where $\hat{\boldsymbol{\sigma}}_t$ is an exponential moving average (EMA) scale estimate with decay β (typically 0.95). Setting radii $r_{t,j} = q_t\hat{\sigma}_{t,j}$ yields heterogeneous prediction sets with *exact* joint coverage. This is not a union bound but a logical identity (cf. Messoudi et al., 2021):

$$P(\forall j : |c_{t,j} - \hat{c}_{t,j}| \le r_{t,j}) = P(s_t \le q_t) \xrightarrow{T\to\infty} 1 - \alpha \tag{3}$$

The identity holds because $\{\forall j : |e_j| \le q\hat{\sigma}_j\}$ is equivalent to $\{\max_j |e_j|/\hat{\sigma}_j \le q\} = \{s_t \le q\}$, reducing a d-dimensional coverage check to a single scalar comparison. The score is dimensionless (invariant to per-component rescaling), and the radii adapt to component-specific magnitudes automatically. This per-component normalization is the multivariate analogue of locally-adaptive conformal prediction (Lei et al., 2018) and conformalized quantile regression (Romano et al., 2019), which scale the nonconformity score by an estimate of local difficulty; here the scale is $\hat{\sigma}_{t,j}$ and the maximum collapses the d components to a single calibrated scalar. This matters when cost components have different units or scales, such as energy prices in \$/MWh alongside start-up costs in \$. Alternative approaches to multivariate conformal coverage include copula-based methods (Cyusa Mukama et al., 2024) that model cross-dimensional dependence explicitly, and dynamic budget allocation (Schlembach et al., 2025) that optimizes how the error budget is distributed across output dimensions. Our max-residual approach trades some tightness for simplicity: a single scalar score suffices for both calibration and optimization.

3.3. Predict-then-Optimize with Conformal Coverage

At each round $t = 1, \ldots, T$ of the online predict-then-optimize pipeline: (1) an ML model (e.g., MLP, LightGBM, or even a simple historical average) produces cost estimate $\hat{\mathbf{c}}_t \in \mathbb{R}^d$; (2) the decision-maker selects $\mathbf{x}_t \in P = \{\mathbf{x} : A\mathbf{x} \le \mathbf{b}, \mathbf{x} \ge 0\}$; (3) true costs $\mathbf{c}_t$ are revealed. No distributional assumptions on $\mathbf{c}_t$ are required.

Given conformal radii $\mathbf{r}_t$, the robust optimization problem is:

$$\min_{\mathbf{x}\in P} \max_{|\mathbf{c}-\hat{\mathbf{c}}|\le\mathbf{r}} \mathbf{c}^\top\mathbf{x} = \min_{\mathbf{x}\in P}(\hat{\mathbf{c}}_t + \mathbf{r}_t)^\top\mathbf{x}$$

where the equality holds for $\mathbf{x} \ge 0$ (the worst case places each cost at its upper bound $\hat{c}_j + r_j$ when $x_j > 0$). This reduces to a single linear program under the perturbed cost $\hat{\mathbf{c}} + \mathbf{r}$. On combinatorial polytopes, classical algorithms apply: Dijkstra for shortest path, Hungarian for assignment, network simplex for minimum-cost flow. Algorithm 1 states the full online loop; the ordering is what keeps it auditable: the decision $\mathbf{x}_t$ is formed from radii $\mathbf{r}_t$ built only from scores observed *before* round t, the true cost $\mathbf{c}_t$ is revealed only afterward, and

Algorithm 1: Online Conformal-Robust Decision (deployment loop)

Input : fixed predictor $\hat{f}$; polytope $P = \{\mathbf{x} : A\mathbf{x} \le \mathbf{b},\, \mathbf{x} \ge 0\}$; target α; ACI rate η; window size W; warm-up W_0; EMA decay β

Output: decisions $\{\mathbf{x}_t\}_{t=1}^{T}$; coverage certificate $\widehat{\mathrm{cov}} = \frac{1}{T}\sum_t \mathbf{1}[\forall j : |c_{t,j} - \hat{c}_{t,j}| \le r_{t,j}]$

```
α₁ ← α;  S ← ∅;  initialize scale σ̂
for t ← 1 to T do
    ĉ_t ← f̂(context_t)                          // predict; f̂ never retrained online
    if |S| < W₀ then
        q_t ← q_def                               // conservative warm-up quantile
    else
        q_t ← Q_{1−α_t}(S)                        // sliding-window quantile, Eq. (1)
    end
    r_{t,j} ← q_t σ̂_{t,j}                          // heterogeneous radii, Eq. (2)
    x_t ← arg min_{x∈P} (ĉ_t + r_t)ᵀ x           // robust LMO (Dijkstra / Hungarian / flow)
    observe c_t                                   // true cost revealed after deciding
    s_t ← max_j |c_{t,j} − ĉ_{t,j}| / σ̂_{t,j}      // normalized max-residual score
    append s_t to S (keep last W);  update EMA scale σ̂ (decay β)
    α_{t+1} ← clip(α_t + η(α − 1[s_t > q_t]), α̲, ᾱ)   // ACI update
end
```

the ACI update (1) then advances the quantile, so no step uses look-ahead information (cf. Figure 2).

A matrix A is *totally unimodular* (TU) if every square submatrix has determinant in $\{-1, 0, +1\}$ (Schrijver, 1998). As integer linear programming is less standard in our community, we state plainly the one consequence we rely on: whenever the right-hand side $\mathbf{b}$ is integer, *every vertex* of the feasible polytope $\{x : Ax \le b,\ x \ge 0\}$ has integer coordinates. Intuitively, TU is the structural reason many combinatorial decisions can be solved as plain linear programs: the LP relaxation never returns a fractional corner, so its optimum is automatically an integral (e.g. 0/1) decision, with no branch-and-bound. The canonical instance is a directed graph's node–arc incidence matrix, which is TU; this is why shortest-path, network-flow, transportation, and bipartite-matching problems have integral LP solutions (Schrijver, 1998), and it is the property shared by our energy-dispatch, taxi-routing, and flu-allocation benchmarks.

Two clarifications prevent a common misreading. First, TU is *not* itself what makes coverage free: any linear program with a unique optimal vertex and a strictly positive cost gap to the next-best vertex is robust to small cost perturbations, TU or not. What TU buys our analysis is that the decision set is *exactly* the integer-feasible set, so the switching threshold κ^* is measured against genuine integral competitors rather than fractional artifacts. Second, a *non-TU* problem is one whose constraint matrix lacks this property: the LP relaxation may then have fractional vertices, and the true integer optimum differs from the LP optimum by a positive *integrality gap*. Facility location is our canonical non-TU case: binary depot open/close variables break total unimodularity and open such a gap. As Proposition 4 and Finding 1 show, the free/costly boundary is governed by the same

Table 1: Totally unimodular (TU) vs. non-TU decision problems. A TU constraint matrix forces every LP vertex to be integral, so the LP relaxation already yields the integer optimum (no *integrality gap*); a non-TU matrix admits fractional vertices, so the integer optimum can differ from the LP optimum. The switching threshold κ^* is defined identically in both classes; only the competing vertices differ (integral vs. fractional).

Problem	Constraint matrix	TU?	Int. gap	In this paper
Shortest path	node–arc incidence	✓	none	taxi, Warcraft
Min-cost flow / transp.	node–arc incidence	✓	none	flu, road networks
Economic dispatch	limits + balance	✓	none	CAISO/ERCOT
Bipartite matching	bipartite incidence	✓	none	—
Facility location	binary + linking	×	positive	facility (NYC)
Knapsack / 0/1 ILP	covering / packing	×	positive	—
Travelling salesman	subtour elimination	×	positive	—

radius-versus-κ^* comparison in both regimes; total unimodularity only determines whether the vertices among which κ^* is measured are themselves integral. Table 1 lists familiar examples of each class and where they appear in our study.

Definition 1 (Decision-neutral coverage) *A conformal prediction set is* decision-neutral *at round t if the robust optimum equals the nominal optimum:*

$$\arg\min_{\mathbf{x}\in P}(\hat{\mathbf{c}}_t + \mathbf{r}_t)^\top \mathbf{x} = \arg\min_{\mathbf{x}\in P} \hat{\mathbf{c}}_t^\top \mathbf{x}.$$

When either minimizer is not unique, we read the definition as: the nominal optimum remains an optimal solution of the robust problem. Throughout the analysis we assume the nominal optimum is unique—equivalently, the cost gap $\Delta(\hat{\mathbf{c}})$ defined below is strictly positive—which holds almost surely when costs are continuously distributed.

When coverage is decision-neutral, the optimizer makes the same decision it would have made without conformal prediction, but the system can additionally certify that "90% of cost realizations fell within our uncertainty sets." This raises a natural question: if coverage never changes the decision, what value does it provide? The first answer reframes the premise. One might object that the purpose of uncertainty quantification is to *alter* decisions; but decision-neutrality is not evidence that the quantification is idle; it is a positive result about the decision itself. A free certificate *proves the chosen action is robust to the calibrated uncertainty*: the hedging a risk-averse operator would otherwise perform is provably unnecessary at the target coverage level, and knowing this, rather than merely assuming it, is actionable. Building on this, we identify three concrete use cases for decision-neutral coverage: (i) *compliance and audit*: regulated industries (energy markets, healthcare) increasingly require documented uncertainty quantification for AI-driven decisions (Renkema et al., 2024); a decision-neutral coverage certificate satisfies this requirement at zero operational cost; (ii) *monitoring*: the coverage rate itself is a canary: if it degrades, the system knows its predictions have deteriorated, even before decisions change; (iii) *assurance*: knowing that coverage is free means the organization can add conformal monitoring without fear of disrupting existing operations, lowering the barrier to adoption.

When coverage is *not* decision-neutral, the prediction set forces a switch to a costlier vertex, and the system pays a premium for its coverage certificate.

4. Main Results

4.1. The Switching Threshold κ^*

Theorem 2 (Decision-Neutral Coverage on TU Polytopes) *Let A be TU with $\mathbf{b}$ integer, let $v^* = \arg\min_{v \in V} \hat{\mathbf{c}}^\top v$ be the unique nominal optimum over the vertex set V, and for $k \neq *$ write $\Delta_k = \hat{\mathbf{c}}^\top v_k - \hat{\mathbf{c}}^\top v^*$, so that $\Delta(\hat{\mathbf{c}}) = \min_{k \neq *} \Delta_k > 0$ is the cost gap to the next-best vertex. If $\max_k |\mathbf{r}^\top (v_k - v^*)| < \Delta(\hat{\mathbf{c}})$, then coverage is decision-neutral: $\mathbf{x}^{\text{robust}} = \mathbf{x}^{\text{nominal}} = v^*$.*

Proof On a TU polytope with integer RHS, every vertex is integral, so the vertices are exactly the integer decisions. For any vertex $v_k \neq v^*$,

$$(\hat{\mathbf{c}} + \mathbf{r})^\top v_k - (\hat{\mathbf{c}} + \mathbf{r})^\top v^* = \Delta_k + \mathbf{r}^\top (v_k - v^*) \;\geq\; \Delta_k - |\mathbf{r}^\top (v_k - v^*)| \;>\; \Delta_k - \Delta(\hat{\mathbf{c}}) \;\geq\; 0,$$

and since a linear program attains its optimum at a vertex, v^* remains the unique robust optimum. For the conformal parameterization $\mathbf{r} = q\boldsymbol{\sigma}$, writing $\delta_k = \sum_j \sigma_j (v_j^* - v_{k,j})$, the comparison above reads $\Delta_k - q\delta_k$: a competitor with $\delta_k \leq 0$ can never overtake v^* under robustification (it uses at least as much noise-weighted capacity as the current optimum), while a competitor with $\delta_k > 0$ ("less noisy") overtakes it exactly at $q = \Delta_k / \delta_k$—the switch points analyzed next. ■

Theorem 3 (Exact Switching Threshold κ^*) *On a TU polytope with $\mathbf{r} = q \cdot \boldsymbol{\sigma}$, define $\delta_k = \sum_j \sigma_j (v_j^* - v_{k,j})$ and $K^+ = \{k \neq * : \delta_k > 0\}$. The* switching threshold *is:*

$$\kappa^* = \min_{k \in K^+} \Delta_k / \delta_k \qquad (\textit{or } \infty \textit{ if } K^+ = \emptyset)$$

*Then: **(i)** coverage is decision-neutral iff $q_t < \kappa_t^*$, where κ_t^* denotes κ^* evaluated at round t's $(\hat{\mathbf{c}}_t, \hat{\boldsymbol{\sigma}}_t)$; the boundary case $q_t = \kappa_t^*$, at which the robust problem has multiple optima including v^*, has measure zero for continuously distributed scores. **(ii)** If $\hat{\sigma}_{t,j} \to 0$ for all j, $\liminf_t \min_{k \neq *} \Delta_k(\hat{\mathbf{c}}_t) > 0$ (e.g., $\hat{\mathbf{c}}_t \to \mathbf{c}$ with a unique limiting optimum), and $\sup_t q_t < \infty$, then $\kappa_t^* \to \infty$ and coverage is eventually always decision-neutral.*

Proof **Part (i).** The relative cost of v_k under robust costs is $\Delta_k - q\delta_k$. For $q < \kappa^*$: competitors with $\delta_k \leq 0$ satisfy $\Delta_k - q\delta_k \geq \Delta_k > 0$, and competitors in K^+ satisfy $\Delta_k - q\delta_k > \Delta_k - \kappa^* \delta_k \geq 0$; hence v^* is the unique robust optimum and coverage is decision-neutral. For $q > \kappa^*$, the minimizing competitor has $\Delta_k - q\delta_k < 0$, so v^* is no longer optimal. This is if and only if, not merely sufficient.

Part (ii). Since there are finitely many vertices, $|\delta_k(t)| \leq \|\hat{\boldsymbol{\sigma}}_t\|_1 \max_j |v_j^* - v_{k,j}| \to 0$ for every k, while the gaps $\Delta_k(\hat{\mathbf{c}}_t)$ stay bounded below by hypothesis, so $\kappa_t^* \to \infty$. As q_t is bounded by hypothesis, $q_t < \kappa_t^*$ for all large t. Each hypothesis matters: predictor consistency drives $\hat{\sigma}_{t,j} \to 0$, but by itself controls neither the drifting gaps $\Delta_k(\hat{\mathbf{c}}_t)$ nor the normalized quantile q_t, which is a ratio of two vanishing quantities. ■

The threshold κ^* is computable via K-shortest-paths or vertex enumeration.[2] Intuitively, κ^* is large when the cost gap Δ_k between the best and second-best vertices is large relative to their noise sensitivity δ_k. On energy dispatch ($d=10$), there are few competing generator commitments and the cost gaps are wide, so κ^* is large: coverage leaves the committed set of generators unchanged and adds no cost. On NYC taxi routing ($d=2{,}308$), many near-optimal alternative paths exist with small cost gaps, so κ^* is small and coverage changes the route on $\sim$50% of rounds.

Proposition 4 (The Switching Mechanism Beyond TU) ***(i)*** *On TU polytopes, there exists a free-coverage region: coverage is decision-neutral for all $q < \kappa^*$ (phase transition at κ^*).* ***(ii)*** *On non-TU polytopes, the characterization of Theorem 3 applies unchanged to the vertices of the LP relaxation, integral or fractional: coverage is decision-neutral iff $q < \kappa^*$, with κ^* now measured between relaxation vertices. Total unimodularity is not what creates the free region; it only guarantees that the competing vertices are integral.*

Proof Part (i) follows from Theorem 3. For part (ii), observe that the proof of Theorem 3 nowhere uses integrality of the vertices: on any polytope, the robust cost of vertex v_k relative to v^* is $\Delta_k - q\delta_k$, so the first switch occurs exactly at $\kappa^* = \min_{k \in K^+} \Delta_k/\delta_k$, the minimum now ranging over all vertices of the relaxation, fractional or integral. The integrality gap $\delta > 0$ determines which nominal the decision-maker holds: if the integer program is solved directly, the same argument applies on the integer-feasible set with κ^* measured between integer vertices, and the two thresholds can differ when the relaxation optimum, or the nearest competing vertices, are fractional. ■

The practical implication is direct: on TU problems (network flow, assignment, transportation), a practitioner can compute κ^* to determine whether coverage is free. On non-TU problems (facility location, lot sizing, scheduling with setup costs), coverage typically has a price at the calibrated radius, and the relevant question becomes how large.

4.2. The Safety Margin: A Pre-Deployment Diagnostic

The switching threshold κ^* determines *where* the phase transition occurs. Whether the conformal quantile q_t exceeds κ^* depends on the stochastic dynamics of the Gibbs-Candès calibration. We characterize these dynamics to provide a pre-deployment diagnostic.

Theorem 5 (Quantile Process Characterization) *Let $\{s_t\}$ be stationary with CDF F, density $f(q^*) > 0$ at $q^* = F^{-1}(1-\alpha)$, and assume the clip in (1) is inactive in steady state ($\alpha \pm O(\sqrt{\eta}) \subset (\underline{\alpha}, \bar{\alpha})$). Under Gibbs-Candès (1) with learning rate η and buffer W:* ***(i)*** $\mathrm{Var}_\infty(\alpha_t) = \eta\alpha(1-\alpha)/2 + O(\eta^2)$. ***(ii)*** *to leading order, treating the α_t-fluctuation and finite-buffer sources as independent, $\sigma_q^2 = \alpha(1-\alpha)/f(q^*)^2 \cdot (\eta/2 + 1/W)$; the neglected cross-covariance between the two sources is of order $\sqrt{\eta/W}$, comparable to the leading terms*

2. On large graphs we compute κ^* from a K-shortest-paths candidate set rather than full vertex enumeration. A K-sensitivity sweep confirms that the resulting predicted decision-neutral fraction $\Pr(q < \kappa^*)$ converges to the candidate-free empirical switching rate as K grows (within 0.6 percentage points at $K=50$); the decision-neutral fractions we report are measured directly from the realized robust-versus-nominal decisions and are therefore independent of this choice.

at practical (η, W), which is one reason σ_q should be estimated empirically in deployment. ***(iii)*** *as $t \to \infty$ with $\eta \to 0$ and $W \to \infty$, $(q_t - q^*)/\sigma_q \xrightarrow{d} \mathcal{N}(0,1)$.*

Proof With the clip inactive, the Gibbs-Candès iteration is an unconstrained constant-step-size stochastic approximation with mean field $H(\alpha_t) = \alpha - \alpha_t$ and unique fixed point $\alpha_t = \alpha$ ($H'(\alpha) = -1$). The noise $\xi_t = \alpha - \mathbf{1}[s_t > q^*]$ is zero-mean Bernoulli with variance $\alpha(1-\alpha)$.

Part (i). By the constant-step-size stochastic approximation central limit theorem (SA-CLT) (Borkar, 2008, Theorem 9.1)(Kushner and Yin, 2003, Ch. 10): $\mathrm{Var}_\infty(\alpha_t) = \eta \cdot \mathrm{Var}(\xi)/(2|H'(\alpha)|) = \eta\alpha(1-\alpha)/2$. The factor $2|H'(\alpha)| = 2$ arises because the drift coefficient -1 provides mean reversion that balances noise injection ($\eta^2\alpha(1-\alpha)$ per step).

Part (ii). Two sources affect q_t. Source 1: α_t fluctuation, via the delta method on $q_t = F^{-1}(1-\alpha_t)$: $\mathrm{Var}_{(1)}(q_t) = \mathrm{Var}(\alpha_t)/f(q^*)^2$. Source 2: finite buffer sampling, with variance $\alpha(1-\alpha)/(Wf(q^*)^2)$ from standard order-statistic theory (Serfling, 1980). The two sources are driven by the same score stream and are not independent; their cross-covariance is of order $\sqrt{\eta/W}$, which is *not* lower-order than $\eta/2+1/W$ (at our operating point $\eta{=}0.05$, $W = 200$ the three quantities are comparable). Summing the two marginal contributions therefore gives the stated *leading-order* formula, with the cross term one contributor—alongside the score autocorrelation treated in the remark below—to the theory–empirics variance gap quantified in Section 5.

Part (iii). Follows from the SA-CLT combined with asymptotic normality of the empirical quantile. ■

Remark (extension to mixing scores). When $\{s_t\}$ is β-mixing (e.g., due to temporal correlation in prediction errors), the SA-CLT still holds (Borkar, 2008, Ch. 5) with modified noise variance $\sigma^2_{\mathrm{mix}} = \alpha(1-\alpha)(1 + 2\sum_{k=1}^{\infty}\rho_k)$, where $\rho_k = \mathrm{Corr}(\mathbf{1}[s_t > q^*], \mathbf{1}[s_{t+k} > q^*])$. This accounts for the $\sim 3\times$ gap between theoretical and empirical σ_q in Section 5.

Theorem 6 (Safety Margin Diagnostic) *On a TU polytope, let G be the marginal distribution of κ^*_t and define the* safety margin $m = (\bar{\kappa}^* - q^*)/\sigma_q$ *with $\bar{\kappa}^* = \mathbb{E}_G[\kappa^*_t]$. Assume* ***(A1)*** *the Gaussian approximation of Theorem 5(iii) for q_t, and* ***(A2)*** *independence of q_t and κ^*_t (both depend on the error process through $\hat{\boldsymbol{\sigma}}_t$; the approximation is validated empirically in Section 5). Then:*

1. *The probability that coverage changes the decision is* $\Pr(q_t > \kappa^*_t) \approx \mathbb{E}_G[\Phi((q^* - \kappa^*_t)/\sigma_q)]$.
2. *When* $\mathrm{Var}_G(\kappa^*) \ll \sigma_q^2$*, this simplifies to* $\Phi(-m)$*, giving three regimes: $m \gg 1$: decision-neutral (*$\Pr = O(e^{-m^2/2})$*); $m \approx 0$: critical (*$\Pr \approx 1/2$*); $m \ll -1$: decision-altering (*$\Pr \to 1$*).*
3. *The safety margin is computable from: (a) the problem structure (κ^*, via vertex enumeration or K-shortest-paths), and (b) empirical estimates of q^* and $\hat{\sigma}_q$ from a short calibration run. The theory (Theorem 5) motivates the Gaussian approximation; the calibration run provides the actual variance, which may exceed the theoretical value due to score autocorrelation.*

Proof The score $s_t = \max_j |c_{t,j} - \hat{c}_{t,j}|/\hat{\sigma}_{t,j}$ depends only on prediction error, not on the decision $\mathbf{x}_t$, so the quantile process $\{q_t\}$ is unaffected by the decisions; independence of

q_t from κ_t^*, however, is genuinely assumption (A2), since both are functionals of the error process through $\hat{\boldsymbol{\sigma}}_t$. Under (A1), $q_t \sim \mathcal{N}(q^*, \sigma_q^2)$ asymptotically; by (A2) and iterated expectation over G: $\Pr(q_t > \kappa_t^*) = \mathbb{E}_G[\Phi((q^* - \kappa_t^*)/\sigma_q)]$. When $\mathrm{Var}_G(\kappa^*) \ll \sigma_q^2$, a first-order expansion of Φ around $\bar{\kappa}^*$ gives $\Phi(-m)$. The three regimes follow from the Gaussian tail bound $\Phi(-m) \le \frac{1}{2}e^{-m^2/2}$ for $m \ge 0$. ■

The safety margin m is the paper's main practical output. A practitioner computes κ^* from the problem structure, then runs a short calibration period to estimate q^* and $\hat{\sigma}_q$ empirically, and evaluates m. A positive m means coverage is likely free; a negative m means it will change decisions on most rounds. The calibration run is necessary because the theoretical σ_q (Theorem 5) assumes independent scores; in practice, temporal autocorrelation inflates the variance by a factor that depends on the score process and must be estimated from data (Section 5). To be precise about what this requires: the diagnostic is pre-*deployment*, not pre-*data*. It consumes a short calibration pilot with the already-trained predictor (a modest, routine step before committing any ML system to production) and reads $\hat{\sigma}_q$ off it directly; the theory fixes the *shape* of the quantile fluctuations (Gaussian), while the pilot supplies their *width*. This still delivers the practical payoff (classifying a problem as free, borderline, or costly *before* the conformal layer is rolled out at scale) without leaning on a theoretical variance the data do not bear out.

4.3. Dimension Dependence

Proposition 7 (Dimension Dependence of κ^*) *On a flow polytope, let the* detour neighbors *of the shortest path P^* be the single-edge detours (replace $(u,v) \in P^*$ by $(u,w),(w,v)$), let K_{det} be their number, and let $K_{\text{det}}^+ = \{k : \delta_k > 0\}$ be those whose original edge is noisier than the detour: **(i)** $\kappa^* \le \min_{k \in K_{\text{det}}^+} \Delta_k/\delta_k$. **(ii)** Under i.i.d. edge costs with density bounded below and i.i.d. nondegenerate noise scales σ_j:* $\mathrm{median}(\kappa^*) = O(1/K_{\text{det}})$ *(proof sketch).* ***(iii)*** *On K_n: $K_{\text{det}} \ge n-2$, so under the assumptions of (ii),* $\mathrm{median}(\kappa^*) = O(1/n)$.

Proof **(i)** Each detour neighbor is a vertex of the flow polytope; since $K_{\text{det}}^+ \subseteq K^+$, $\kappa^* \le \min_{K_{\text{det}}^+} \Delta_k/\delta_k$. For a single-detour through w replacing edge $(u,v) \in P^*$ with $(u,w),(w,v)$: $\delta_k = \sigma_{(u,v)} - \sigma_{(u,w)} - \sigma_{(w,v)}$, positive when the original edge is noisier than the detour. **(ii)** (Sketch.) Nondegenerate i.i.d. scales give each detour probability $p_0 > 0$ of $\delta_k \ge \delta_0 > 0$, so $|K_{\text{det}}^+|$ grows in proportion to K_{det}; heterogeneous scales are essential—with identical scales every single-edge detour has $\delta_k = -\sigma < 0$ and K_{det}^+ is empty. Group detours by the replaced edge. Under i.i.d. costs with density $f \ge f_0 > 0$, the minimum of the $\sim p_0 K_{\text{det}}$ gaps with $\delta_k \ge \delta_0$ is $O(1/K_{\text{det}})$ with probability at least 1/2 (order statistics; the gaps share the cost of P^* and are treated as near-independent, which is where this is a sketch), whence $\mathrm{median}(\kappa^*) \le O(1/K_{\text{det}})/\delta_0$. We state the bound for the median—the quantity our experiments report—rather than the mean, since $K_{\text{det}}^+ = \emptyset$ with positive probability on small graphs, making $\mathbb{E}[\kappa^*]$ infinite. **(iii)** On K_n, each $w \notin \{s,t\}$ gives a 2-hop detour, so $K_{\text{det}} \ge n-2$. ■

The scaling is intuitive: denser graphs have more near-optimal alternative paths, shrinking the cost gap to the nearest competitor and thus shrinking the decision-neutral region.

This explains the full spectrum: energy dispatch with 10 generators has few alternatives (large κ^*, free coverage), while taxi routing on a 50-zone graph has $\sim$50 near-optimal detours (small κ^*, coverage changes decisions frequently).

5. Experiments

We validate the switching threshold and safety margin on four real-world benchmarks (Table 2). Each benchmark was selected to probe a specific aspect of our theory: low-dimensional TU (energy), medium-dimensional TU with shift (flu), high-dimensional TU (taxi), and non-TU (facility). All use normalized max-residual conformal scoring (2) with $\alpha = 0.1$, $\eta = 0.05$, $W = 200$, and 3–5 random seeds. These four core benchmarks run on a single Apple M2 Pro CPU in <30 minutes total; two further case studies then extend the validation to metropolitan-scale road networks under equilibrium congestion (Section 5.5) and to a vision-based routing benchmark (Section 5.6).

5.1. Benchmark Descriptions

CAISO energy dispatch ($d = 10$, TU). We use 8,760 real hourly locational marginal prices (LMP) from the California Independent System Operator's OASIS portal (California Independent System Operator, 2023) for 2023, across three pricing zones (SP15, NP15, ZP26). Prices average \$40–60/MWh with spikes to \$1,247/MWh and 146 negative-price hours. An MLP predictor trained on January–June is evaluated over the 185-day July–December period with natural seasonal demand shift. The decision problem is a 10-generator economic dispatch (TU). We chose this benchmark as a low-dimensional TU polytope where we expect κ^* to be large. The price spikes and seasonal shift test whether decision-neutrality holds even under significant nonstationarity.

NYC taxi routing ($d = 2{,}308$, TU). We construct a shortest-path routing problem from 5.7 million real NYC Taxi and Limousine Commission (TLC) trip records (New York City Taxi and Limousine Commission, 2023) (January and July 2023). The city is partitioned into 50 zones, yielding a directed graph with 2,308 edges whose costs are hourly mean travel times. A LightGBM predictor trained on January data is evaluated on July data (seasonal congestion shift), with decisions solved via Dijkstra's algorithm. We chose this benchmark because Proposition 7 predicts that κ^* will be small on high-dimensional TU polytopes with many detour alternatives. It probes the regime where coverage is *not* free despite the TU property.

CDC flu antiviral allocation ($d = 50$, TU). We use 4,680 real weekly influenza-like illness (ILI) rate observations from CDC FluView (Centers for Disease Control and Prevention, 2023) (2015–2023) across 10 HHS regions. The decision is a minimum-cost transportation LP allocating antivirals from 5 supply depots to 10 demand regions ($d = 50$ decision variables, TU constraint matrix). The predictor is trained on pre-pandemic data (2015–2020) and evaluated on 2021–2023, which includes the post-COVID ILI resurgence (7.5% ILI vs. typical 3–4%). We selected this benchmark specifically to test whether decision-neutrality survives a genuine public health crisis that dramatically changes the cost landscape.

Table 2: Summary of real-world benchmarks. All distribution shifts are natural (seasonal or temporal), not synthetic. Each benchmark was selected to probe a specific theoretical prediction.

Dataset	d	TU?	Records	Shift	Probes
CAISO Energy	10	≈✓	8,760 hrs	Seasonal demand	Low-d TU: free?
NYC Taxi	2,308	✓	5.7M trips	Seasonal congestion	High-d TU: costly?
CDC FluView	50	✓	4,680 wk-rgn	Post-COVID	Shift robustness
Facility (NYC)	260	×	—	Seasonal	Non-TU: always costly?

Table 3: Decision-neutrality across benchmarks (3–5 seeds). "κ^* frac." is the fraction of rounds where $q_t < \kappa_t^*$ (decision-neutral). "Decision change" is the fraction where the robust and nominal decisions differ. For energy dispatch the decision is the committed set of generators (which coverage leaves unchanged, adjusting only the marginal generator's output at zero price of coverage); for the routing and facility problems it is the combinatorial solution itself (path, allocation, or open-depot set). All four benchmarks behave as the theory predicts.

Problem	d	TU?	κ^* frac.	Decision change	Coverage
Energy (CAISO)	10	≈✓	>99%	<1%	89%
Flu (CDC)	50	✓	93%	5%	91%
NYC Taxi	2,308	✓	52%	48%	90%
Facility (NYC)	260	×	33%	67%	90%

NYC taxi facility location ($d = 260$, non-TU). A capacitated facility location problem using real taxi data (New York City Taxi and Limousine Commission, 2023) for both transport costs and demand, with 25 zones, 10 candidate depot locations, and $d = 260$ decision variables. Linking constraints (binary depot open/close variables) break the TU property, yielding a 0.97% integrality gap. This benchmark serves as a control: Proposition 4(ii) predicts a positive price of coverage on non-TU polytopes, testing the structural separation between TU and non-TU regimes.

5.2. Decision-Neutrality Across Benchmarks

Finding 1: Decision-neutrality depends on dimension, not TU alone. Table 3 shows the central result. On CAISO energy ($d = 10$), coverage is *free*: it never changes which generators are committed, and the price of coverage is essentially zero: the robust and nominal schedules cost the same to within measurement noise. The 10-generator dispatch has few competing commitments and wide cost gaps (κ^* is large relative to q_t), so the calibrated uncertainty set only adjusts the marginal generator's output, at no cost, while leaving the committed set of generators intact. The system gets a calibrated 89% coverage certificate entirely for free.

On CDC flu ($d=50$), decision-neutrality holds on 93% of rounds despite the post-COVID shift (7.5% ILI vs. typical 3–4%). The 50-dimensional transportation LP has moderate cost

Table 4: Safety margin validation across NYC taxi dimension sweep (3 seeds). The safety margin m correctly identifies the operating regime on every graph size. The theoretical σ_q underestimates by $\sim 3\times$ due to score autocorrelation; the empirical $\hat{\sigma}_q$ resolves this.

n	d	q^*	$\hat{\sigma}_q$	$\bar{\kappa}^*$	m	Regime
10	90	4.8	1.45	16.5	+8.1	decision-neutral
20	380	6.2	2.21	11.3	+2.3	decision-neutral
30	870	6.7	2.31	8.5	+0.8	critical
40	1,555	8.8	3.03	4.5	−1.4	decision-altering
50	2,308	10.0	3.19	4.2	−1.8	decision-altering

gaps, and even the severe shift does not push q_t above κ^* on most rounds. Coverage is 91%, close to the 90% target, confirming that the Gibbs-Candès update adapts to the shift. The 7% of rounds where coverage changes the decision correspond to extreme ILI spikes in specific regions, where the conformal radii are large enough to reroute antiviral shipments.

On NYC taxi ($d=2{,}308$), coverage changes the decision on 48% of rounds. The graph has $K_{\text{det}} \approx 50$ near-optimal detour paths, which shrinks κ^* enough that the conformal quantile frequently exceeds it. As we show in Finding 4, halving prediction error barely changes this fraction, confirming that graph topology rather than prediction quality drives the switching. Coverage is maintained at 90%, but at the cost of routing traffic on longer, less-noisy paths.

On facility location (non-TU, $d=260$, 0.97% integrality gap), coverage changes the LP-relaxation decision on 67% of rounds (3 seeds). A per-round decomposition attributes these switches to the radius-versus-gap mechanism of Proposition 4(ii) rather than to fractional optima: under the (smooth) predicted costs the LP-relaxation optimum is integral on every measured round, and every switch is a genuine depot-pattern change between integer vertices whose cost gap the calibrated radius exceeds; fractional optima arise only under the noisier realized costs ($\sim$17% of rounds), which is where the measured 0.97% gap lives. Scaling the conformal radius by a factor λ traces the free region directly: the integer depot pattern is unchanged on 100% of rounds at $\lambda = 0$, 50% at $\lambda = 0.1$, and 33% at the calibrated radius, with a price of coverage of only +0.6%. The switching threshold κ^* of Theorem 3, now the gap between *integer* vertices, governs this region exactly as on TU polytopes. Thus the free/costly distinction is not fundamentally TU versus non-TU, but whether the conformal radius stays below the switching gap of whichever nominal (fractional or integer) the decision-maker uses.

5.3. Safety Margin Validation

Finding 2: The safety margin correctly classifies every instance. To validate the safety margin as a pre-deployment diagnostic, we sweep the NYC taxi graph from $n = 10$ to $n = 50$ zones (Table 4, Figure 3). This creates a controlled sequence of problems with increasing dimension and decreasing κ^*.

At $n=10$ ($d=90$), the safety margin $m = +8.1$ is deeply positive, and coverage changes the decision on fewer than 9% of rounds. At $n=30$ ($d=870$), the margin drops to $m = +0.8$,

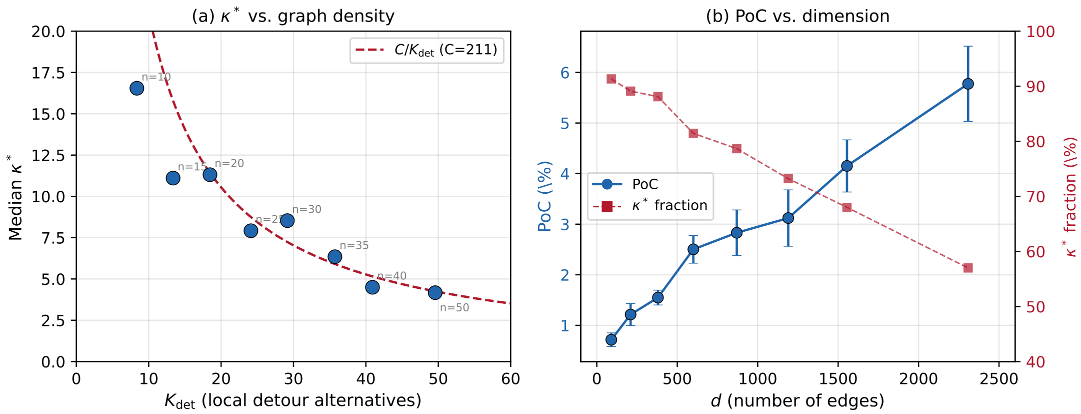

Figure 3: **Left:** Median κ^* vs. K_{det}; dashed line is the fitted C/K_{det} curve (Proposition 7). **Right:** Price of Coverage (blue, left axis) and the decision-neutral fraction $\Pr(q_t < \kappa_t^*)$ (red, right axis) vs. d. As graph density grows, κ^* shrinks: the Price of Coverage rises while the decision-neutral fraction falls, and coverage changes the route more often, at higher cost. The transition from decision-neutral to decision-altering occurs around $n = 30$ ($d \approx 870$), consistent with the critical regime at $m \approx 0.8$.

near the critical boundary, and coverage changes the decision on about 21% of rounds. At $n=50$ ($d=2{,}308$), $m = -1.8$ is solidly negative, and 43% of rounds see a decision change. All three regimes predicted by Theorem 6 are visible in the data.

One caveat: the theoretical σ_q from Theorem 5(ii) underestimates the empirical $\hat{\sigma}_q$ by roughly 3×. The conformal scores $\{s_t\}$ exhibit temporal autocorrelation because spatially correlated prediction errors on neighboring edges persist across rounds, inflating the effective variance via the mixing correction $\sigma^2_{\text{mix}} = \alpha(1-\alpha)(1 + 2\sum_k \rho_k)$. Using $\hat{\sigma}_q$ in the safety margin formula resolves this gap and produces correct regime classifications across all graph sizes.

5.4. Dimension Dependence Validation

Finding 3: κ^* scales as $O(1/K_{\text{det}})$. Table 5 validates Proposition 7 on real data. As d grows from 90 to 2,308, the number of detour alternatives K_{det} scales linearly with n ($K_{\text{det}} \approx n$), the median κ^* decreases monotonically, and the decision-neutral fraction drops from 91% to 57%. The product $\kappa^* \cdot K_{\text{det}}$ ranges from 138 to 249, approximately constant for $n \geq 15$, consistent with $O(1/K_{\text{det}})$. The elevated variance at $n = 10$ ($K_{\text{det}} = 8.4$, product 138) is expected: the minimum of 8 gap ratios has higher relative variance than the minimum of 50. This scaling is robust to how detour neighbors are defined: under alternative definitions (unfiltered two-hop detours, and a path-based count of near-shortest paths within a $(1{+}\epsilon)$ cost margin), the alternative counts remain proportional to the edge-local K_{det} (the path-based count is 0.09–0.14 × K_{det} across dimensions) and the $O(1/K_{\text{det}})$

Table 5: Dimension sweep on NYC taxi flow polytopes (3 seeds). The product $\kappa^* \cdot K_{\text{det}}$ is approximately constant (range 138–249), consistent with the $O(1/K_{\text{det}})$ scaling of Proposition 7. Variance at small n reflects order-statistic effects.

n	d	K_{det}	$\tilde{\kappa}^*$	κ^* frac.	$\tilde{\kappa}^* \cdot K$
10	90	8.4	16.5	91%	138
15	210	13.4	11.1	89%	148
20	380	18.5	11.3	88%	209
25	600	24.1	7.9	81%	191
30	870	29.2	8.5	79%	249
35	1,190	35.7	6.3	73%	227
40	1,555	40.9	4.5	68%	184
50	2,308	49.6	4.2	57%	207

fit is preserved, confirming that K_{det} is a structural statistic rather than an artifact of the specific detour definition.

The physical interpretation is straightforward. On a sparse graph ($n = 10$), the shortest path between most origin-destination pairs is nearly unique, so the cost gap to the next-best path is large. On a dense graph ($n = 50$), many near-optimal paths exist, cost gaps shrink, and conformal radii more easily flip the optimal route. The number of detour alternatives, not dimensionality per se, is the fundamental quantity.

Finding 4: Decision-neutrality is structural, not a predictor artifact. A predictor ablation at $d{=}2{,}308$ rules out the possibility that decision-neutrality failures are caused by poor prediction quality. Reducing MAPE by 35% (from 20.5% with a naive January-mean predictor to 13.3% with LightGBM) barely changes the κ^*-fraction (60% $\rightarrow$ 52%). Even a perfect predictor would face the same graph topology: $\sim$50 near-optimal detour paths create small cost gaps that conformal radii easily exceed. Problem structure, not ML model quality, is the binding constraint.

The practical implication is that investing in a better predictor will not make coverage free on high-dimensional problems. The path to free coverage on dense graphs is structural: simplifying the graph (fewer zones), not improving predictions.

5.5. Case Study: The Price of Coverage in Congested Road Networks

Background: user equilibrium and the BPR link function. Traffic assignment is a canonical network-optimization problem of transportation science, and one increasingly cast as predict-then-optimize as agencies learn its inputs from data. Given a road network and an origin–destination (OD) travel-demand matrix, the network's operating state is the *user equilibrium* (UE): the flow pattern in which no traveler can lower their own travel time by unilaterally switching routes (Wardrop, 1952). Because a link's travel time rises with its flow (congestion), the UE is not a plain shortest-path assignment; Beckmann et al. (1956) showed it is the unique minimizer of a convex program (the Beckmann transformation) whose objective integrates each link's travel-time function. The standard link travel-time

Table 6: Five real road networks under User-Equilibrium (BPR) congestion, each validated against the published TNTP reference flows (link-flow correlation ≥ 0.999, total-travel-time error $< 0.1\%$). Conformal coverage stays calibrated near the 90% target at every scale, from $d = 76$ to $d = 40{,}003$ edges; decision-neutrality is substantial throughout but is set by the switching gap κ^* (which shrinks as more routes compete on larger networks), not by d, K_{det}, or congestion alone. Mean $\pm$ std over 3 seeds.

Network	Edges d	K_{det}	Decision-neutral	Coverage
Sioux Falls	76	0.3	91% ±1	91%
Anaheim	914	3.1	84% ±2	91%
Chicago-Sketch	2,950	0.8	85% ±1	91%
Chicago-Regional	39,018	1.2	59% ±2	91%
Philadelphia	40,003	0.4	71% ±1	92%

function is the Bureau of Public Roads (BPR) form

$$t_a(x_a) = t_a^0 \left[1 + \alpha \, (x_a/c_a)^\beta \right], \tag{4}$$

with free-flow time t_a^0, capacity c_a, and standard constants $\alpha = 0.15$, $\beta = 4$ (Bureau of Public Roads, 1964), which we read per link from the benchmark files; it is the de-facto standard in planning practice and is the cost model published with every benchmark network we use. BPR congestion is what makes the problem non-trivial for our purposes: under free-flow costs, link times are constant and routing decomposes into independent shortest paths, whereas under BPR the equilibrium travel times are *flow-dependent*, so the very cost vector a downstream router optimizes against is itself an output of the equilibrium. We solve the Beckmann program with the Frank–Wolfe algorithm (Frank and Wolfe, 1956): at each iteration we linearize the objective, solve an all-or-nothing shortest-path subproblem, and take an exact line-search step toward it. This is the classical traffic-assignment method of LeBlanc et al. (1975) and remains a textbook standard (Sheffi, 1985). Frank–Wolfe is the natural solver here for the same reason the linear minimization oracle is central to our decisions elsewhere: a shortest-path assignment is cheap, while projection onto the flow polytope is not. Figure 4 visualizes one such equilibrium on the Sioux Falls network: some links carry travel times several times their free-flow value, and it is these congested times, not the free-flow times, that the downstream router hedges against. We validate every computed equilibrium against the published reference flows before running any conformal analysis (Table 6).

Finding 5: Coverage calibration is scale-invariant under real congestion. The dimension sweep above varies a single graph family under free-flow costs. To test whether the coverage certificate survives across *heterogeneous* real networks at much larger scale and under genuine congestion, we run the online conformal loop on the validated equilibrium costs of five real road networks from the Transportation Networks benchmark (Transportation Networks for Research Core Team, 2024) (Table 6), spanning Sioux Falls ($d = 76$) to Philadelphia ($d = 40{,}003$). Conformal coverage stays within one to two points of the 90%

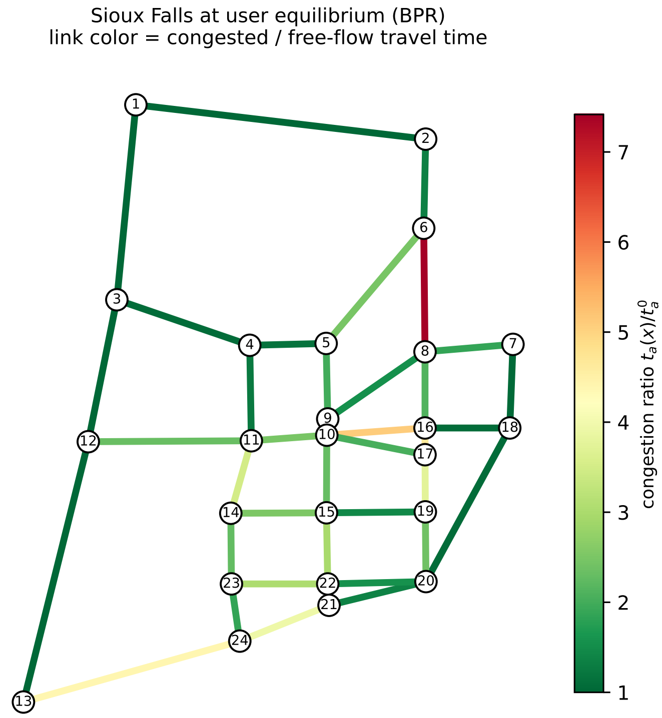

Figure 4: The canonical Sioux Falls network drawn at its real geographic node coordinates, solved to User Equilibrium under the BPR link function. Each link is colored by its congestion ratio $t_a(x)/t_a^0$ (congested over free-flow travel time); the busiest links carry over 7× their free-flow time. The equilibrium travel times are flow-dependent, so the cost vector the conformal layer robustifies is itself an output of the assignment, the setting of our transportation case study.

target on *every* network across a 525× range in dimension: high dimension does not degrade the coverage certificate, and the auditability guarantee holds intact at $d = 40{,}003$. Decision-neutrality remains substantial throughout (59–91%), but across these heterogeneous networks it is governed by the switching gap κ^* (the cost gap to the nearest competing route) rather than by d or the local detour count K_{det} in isolation. Neither alone predicts it: the two largest networks are of similar size yet differ markedly (Chicago-Regional 59% vs Philadelphia 71% at $d \approx 40{,}000$), and the small but locally dense Anaheim grid ($K_{\text{det}} = 3.1$, 914 links) is *more* decision-neutral (84%) than the far larger, locally sparse Philadelphia ($K_{\text{det}} = 0.4$, 40,003 links). What consistently governs neutrality is κ^*: small, sparse networks admit few competing routes and keep κ^* large, whereas larger networks accumulate many near-cost-tied routes that shrink it. Congestion is not the driver: the most congested network (Sioux Falls, 2.3× free-flow) is the most decision-neutral because it is the smallest. This is the same radius-versus-κ^* mechanism as in the single-family sweep (Table 5), now confirmed across heterogeneous real networks an order of magnitude larger.

Why the switching threshold matters for transportation planning. Transportation agencies increasingly derive OD and travel-time inputs from machine-learned predictions (loop detectors, probe vehicles, or camera feeds) that feed the same equilibrium-and-routing pipeline. Routing on these congested networks is itself an active optimization problem: Mantri et al. (2024), for instance, reroute unoccupied (*deadheading*) autonomous vehicles onto longer paths through a mixed user-equilibrium/system-optimum assignment, on the same BPR benchmark networks we study (including Sioux Falls), to limit their congestion contribution while still meeting household mobility needs. Such assignments take the congestion-dependent travel-time vector as *given*; once that vector is instead learned from data, the same decision inherits the predict-then-optimize structure our diagnostic targets. Conformal prediction offers these agencies a distribution-free way to attach calibrated uncertainty to those learned inputs, which raises the operational question this paper answers: would maintaining that coverage actually change the assignment they deploy, or merely document it? Our results give a deployable answer on real metropolitan networks. First, the coverage certificate is *scale-invariant*: it holds at 90–92% on the full Chicago-Regional and Philadelphia networks ($d \approx 40{,}000$ links), so an agency can audit calibrated uncertainty over an entire metropolitan network without the guarantee eroding. Second, whether that coverage is *free* or *costly* is decided before deployment by the switching threshold κ^*: on small, sparse networks few routes compete, κ^* is large, and coverage is nearly free (Sioux Falls, 91% decision-neutral), whereas on large, dense networks many near-cost-tied routes coexist, κ^* shrinks, and coverage reroutes a substantial share of traffic (Chicago-Regional, 41% of rounds). As Finding 5 showed, this boundary is set by network *size*, not congestion; heavier congestion alone does not make coverage costly. A planner can therefore compute κ^* from the network and a short calibration run (using standard traffic-assignment machinery) and know, *in advance*, whether conformal coverage will leave the assignment unchanged (and simply provide a compliance-ready audit trail at no cost) or actively reroute flows (and must be budgeted for). This is precisely the pre-deployment call our safety-margin diagnostic (Table 4) is designed to make, now demonstrated on the field's own canonical equilibrium model at city scale.

5.6. A Vision Modality: Terrain Routing on Warcraft II

Every benchmark so far pairs a *tabular* predictor with the optimizer. A natural stress test is whether the switching-threshold analysis survives when the predictor is a deep vision model whose errors are spatially structured rather than independent across cost components. We therefore apply the framework to the canonical vision predict-then-optimize benchmark of Vlastelica et al. (2020). Each instance is a real 96×96 RGB image of Warcraft II terrain; a convolutional network predicts the underlying 12×12 grid of per-tile traversal costs, and the decision is the minimum-cost, 8-connected path from the top-left to the bottom-right tile. This is a totally-unimodular shortest-path polytope of dimension $d = 144$, so it inherits the same theory as our taxi benchmark, but the cost vector is now the output of image perception. Figure 5 shows one real instance: from pixels alone the network must infer

Figure 5: A real Warcraft II routing instance from the ICLR-2020 benchmark (Vlastelica et al., 2020). **(a)** the 96×96 RGB terrain image the CNN sees; **(b)** the true 12×12 tile costs it must predict (rock/water ≈ 8–9, dirt ≈ 1); **(c)** the resulting minimum-cost 8-connected path, which routes around the expensive rock and water. The many equal-cost detours through cheap terrain make the switching threshold κ^* small, so coverage frequently reroutes the path, but between near-cost-tied alternatives, at negligible price.

that rock and water tiles are expensive and dirt is cheap, and the optimal path threads the low-cost terrain.[3]

On the held-out real images the CNN is accurate (mean absolute percentage error 4.4%) and conformal coverage is calibrated at $88.6\% \pm 0.3$, confirming that the coverage guarantee of the normalized max-residual score is unaffected by the spatially correlated perception errors, as expected, since that score reduces joint coverage to a single exchangeable scalar regardless of the error geometry. The calibrated set changes the route on 64% of rounds (naïvely a "costly" regime), yet the *realized* cost premium is only $0.19\% \pm 0.02$ of the optimal path cost. This apparent tension sharpens our thesis and exposes a distinction the tabular benchmarks do not surface. Whether coverage *changes* the decision is governed by κ^*, which is small here because a terrain grid (Figure 5) admits many near-cost-tied paths through cheap tiles; but the *price* of a change is governed by the cost gap actually traversed, which is negligible precisely when the competing routes are near-tied. Decision-change and price of coverage are therefore distinct: a small κ^* makes coverage decision-*altering* without making it *expensive*. Operationally this is a benign regime: the certificate is auditable at essentially no cost, even though the raw decision-change fraction is high. (We report the absolute premium rather than the ratio PoC here because a near-optimal predictor drives the SAA regret toward zero, making the ratio ill-conditioned.)

3. Real images and terrain-cost labels from the published benchmark (Vlastelica et al., 2020), obtained via the redistributed `npz` files at https://github.com/archettialberto/tilebased_navigation_datasets. A CNN (three convolutional blocks with batch normalization) is trained on 10,000 real training images; we report mean $\pm$ std over 3 seeds on the held-out test set.

6. Conclusion

We introduced the switching threshold κ^*, the first quantile value at which conformal coverage changes the downstream decision in an AI-driven optimization pipeline. On TU polytopes, decision-neutrality holds iff $q < \kappa^*$. The safety margin $m = (\kappa^* - q^*)/\hat{\sigma}_q$, estimated from a calibration run, predicts the operating regime, and the dimension dependence $\kappa^* = O(1/K_{\text{det}})$ explains the transition from free to costly coverage as problem complexity grows. Four real-world benchmarks validate all predictions: energy dispatch (free), flu allocation (free despite severe shift), taxi routing (costly due to dimension), and facility location (costly at the calibrated radius, governed by the same switching-gap mechanism). More broadly, our results indicate that the free/costly boundary is not fundamentally about total unimodularity: TU guarantees that the optimum is an integral vertex, but coverage is free on any polytope (TU or not) precisely when the conformal radius stays below the switching gap κ^* to the nearest competing vertex. The facility optimum, like the TU benchmarks, exhibits this free region: its switches are radius-exceeding-gap events between integer vertices, not artifacts of fractional relaxation optima. A separate study on five real road networks under validated User-Equilibrium congestion confirms that the coverage certificate itself stays calibrated at scales up to $d = 40{,}003$, with decision-neutrality set by κ^* rather than dimension (Table 6). A vision case study (routing on real Warcraft II terrain images) further separates two notions the tabular benchmarks conflate: a small κ^* makes coverage decision-*altering* (it reroutes 64% of paths) yet essentially *free* (a 0.19% cost premium), because the competing routes it switches among are near-cost-tied.

6.1. Practical Guidance

For any LP-based AI decision system with conformal coverage:

1. Compute κ^* from the problem structure (via K-shortest-paths or vertex enumeration). This step requires only the constraint matrix and a representative cost vector, not deployment data.
2. Run a short calibration period (e.g., 200–500 rounds) to estimate q^* and $\hat{\sigma}_q$ empirically. The calibration must use the deployed predictor and score function.
3. Compute $m = (\kappa^* - q^*)/\hat{\sigma}_q$.
4. If $m > 2$, coverage is free; deploy with confidence. If $m < -2$, coverage will change decisions on most rounds; budget for the cost. If $|m| < 2$, the system is in the critical regime and should be monitored closely.

Algorithm 2 (Appendix A) states this diagnostic as pseudocode.[4]

6.2. Limitations and Future Work

1. **Uncertainty set geometry.** Our analysis assumes the axis-aligned box set produced by the normalized max-residual score. Cost components in many domains (e.g., travel times on adjacent edges, prices across zones) are strongly correlated, making the box

4. An open-source implementation is available in the `conformal-ops` package (https://github.com/chandrad/conformal-ops; `pip install conformal-ops`): the diagnostic of Algorithm 2 ships as `FreeCoverageDiagnostic`, and the online conformal–robust loop of Algorithm 1 reuses its `OnlineConformal` calibrator.

conservative relative to the true error geometry. Correlation-aware sets, such as copula-based methods (Cyusa Mukama et al., 2024; Messoudi et al., 2021) or ellipsoidal sets, would produce tighter uncertainty regions that bind differently, potentially shifting the free/costly classification. All results in this paper are conditional on the box geometry; extending the κ^* analysis to non-axis-aligned sets is an important open direction.

2. **ACI vs. per-step coverage.** The ACI update provides long-run average coverage, not per-step guarantees; during distribution shifts, coverage may dip for $O(1/\eta)$ rounds before the update adapts. The regime classification is most reliable during stationary periods and should be interpreted with caution during shift episodes.
3. The κ^* diagnostic applies to LP-representable problems; extension to nonlinear or mixed-integer objectives is an important open direction.
4. **Energy dispatch is only approximately TU.** Real economic dispatch with ramp-rate and operating-reserve constraints couples decisions across time and breaks exact total unimodularity, so on such instances the integral-vertex property (and the tight κ^* characterization of Theorem 3) holds only approximately, placing them under the non-TU case of Proposition 4.
5. The theoretical σ_q underestimates empirical fluctuations by $\sim 3\times$ due to score autocorrelation; the mixing extension (Theorem 5) provides the correction, but requires estimating the autocorrelation structure from a calibration run.
6. Our controlled dimension sweep (Table 5) reaches $d = 2{,}308$, and the road-network study (Table 6) extends coverage validation to $d = 40{,}003$ under real equilibrium congestion; problems with even larger implicit polytopes (e.g., continental vehicle routing with 10^6 edges) remain open.
7. The connection between κ^* and *conditional* coverage (Gibbs et al., 2025) (per-instance rather than marginal) is unexplored.

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Appendix A. Pre-Deployment Free-Coverage Diagnostic

Algorithm 2 formalizes the safety-margin triage described in Section 5's Safety Margin Validation and the Practical Guidance of the Conclusion. It composes the structural threshold κ^* (computable from the constraint matrix alone) with a short calibration pilot that estimates the quantile process $(\hat{q}^*, \hat{\sigma}_q)$, and returns the free/critical/costly regime via the margin $m = (\kappa^* - \hat{q}^*)/\hat{\sigma}_q$. The pilot reuses the online loop of Algorithm 1, so the diagnostic is *pre-deployment* but not *pre-data*: it needs the already-trained predictor and a few hundred calibration rounds, not the full deployment stream.

Algorithm 2: Pre-Deployment Free-Coverage Diagnostic

Input : constraint matrix A, RHS $\mathbf{b}$; representative cost $\hat{\mathbf{c}}$; deployed predictor $\hat{f}$ and score; target α; calibration length N; candidate count K

Output: regime $\in \{$FREE, CRITICAL, COSTLY$\}$ and margin m

```
κ* ← min_{k∈K+} Δ_k/δ_k over the K shortest-path (or enumerated) competitors of
  arg min_{x∈P} ĉ^⊤ x                      // structure only; no deployment data
run Algorithm 1 for N rounds with (f̂, α); collect the quantile trace {q_t}_{t=1}^N
q̂* ← mean({q_t});   σ̂_q ← std({q_t})          // empirical width from the pilot
m ← (κ* − q̂*)/σ̂_q
if m > 2 then
    return FREE, m                  // coverage decision-neutral w.h.p.; deploy
else if m < −2 then
    return COSTLY, m            // coverage changes most decisions; budget the
      premium
else
    return CRITICAL, m                  // borderline; monitor coverage closely
end
```